

PROBLEM

Problem 1: Water Jug Problem

You are given two jugs, one with a capacity of 4 gallons and the other with a capacity of 3 gallons. Neither jug has any measuring marks. A pump can be used to fill the jugs with water. Water can be poured from one jug to another or emptied onto the ground. There are no other measuring devices available.

Objective: Obtain exactly 2 gallons of water in the 4-gallon jug using only the allowed operations.

Assumptions:

- A jug can be filled completely from the pump.
- Water can be emptied onto the ground.
- Water can be transferred from one jug to another until one jug becomes full or the other becomes empty.
- No additional measuring devices or containers are available.

Problem 2: Mars Rover Intelligent Agent

A Mars Rover is an autonomous robotic vehicle designed to explore the surface of Mars. It must safely navigate through rocky terrain, avoid obstacles, collect scientific samples, monitor its battery level, and communicate with mission control on Earth. The rover uses sensors such as cameras, proximity sensors, temperature sensors, and battery monitors to perceive its environment. Based on the sensor data, it decides whether to move, stop, collect samples, recharge using solar panels, or transmit data back to Earth.

Objective: Analyze the Mars Rover as an intelligent agent by identifying its PEAS (Performance Measure, Environment, Actuators, and Sensors), characterizing its operating environment, determining the most suitable intelligent agent model, explaining how its performance is evaluated, and justifying the appropriate agent architecture.

Problem 3: 8-Queens Problem

The 8-Queens Problem is a classic Artificial Intelligence search problem in which 8 queens must be placed on an 8×8 chessboard so that no two queens attack each other. A queen can attack another queen if they are in the same row, column, or diagonal.

The given board configuration (shown in the question) is not a valid solution because the queen in the first column is on the same diagonal as the queen in the last column.

Objective: Formulate the 8-Queens problem by identifying its initial state, state space, actions (operators), transition model, goal test, path cost, and the appropriate search strategy to obtain a valid arrangement of eight non-attacking queens.

Problem 4: OLA Cab Booking Intelligent Agent

An OLA Cab Booking System is an intelligent application that helps customers book a cab to travel from a source location to a destination. The system considers factors such as cab availability, travel distance, estimated fare, traffic conditions, and customer preferences before selecting the most suitable cab. The system should make intelligent decisions to provide an efficient and convenient booking experience.

Objective: Design the OLA Cab Booking System as an intelligent agent by identifying the appropriate agent model, explaining how it selects the best cab based on the customer's requirements, and writing the pseudocode for the cab booking process.

Problem 5: Uniform Cost Search (UCS)

A logistics company operates a delivery network where packages are transported between warehouses. Each route between warehouses has an associated travel cost representing fuel and transportation expenses.

The delivery network is represented by the weighted graph provided in the question.

Objective: Apply the Uniform Cost Search (UCS) algorithm to find the least-cost path from the start node (S) to the goal node (G). Show the step-by-step node expansion, calculate the total path cost, draw the search tree, and identify the optimal path based on the given graph.